

Financial micro-incentives and internet discourse: Evidence from a Bitcoin based discussion platform*

Edward Kung	Keyan Kousha
CSUN	Stacker News
edward.kung@csun.edu	kk@stacker.news

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Abstract

We study the role of financial micro-incentives in driving user behavior on a Bitcoin-based discussion platform. On the platform, users must pay Bitcoin to post but can also earn tips from other users for posting good content. We find three main results. First, when the cost to post in a subforum increases, the number of posts in that subforum decreases but the quality of the posts increases. Second, the incentive to earn tips leads to higher post quality over time as users learn what content others value. Third, unprofitable users are more likely to leave the platform while profitable users are more likely to stay. The findings suggest that financial micro-incentives may be effective tools to improve the quality of discourse in online discussion platforms. Moreover, the incentives present on the platform are only possible because of Lightning Network’s ability to instantaneously process small, global micro-payments, thus highlighting an important use case of Bitcoin.

Keywords: social media, bitcoin, market design, signaling, learning, knowledge production

JEL Classification: D82, D83, L82

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1 Introduction

Public discourse for matters both mundane and substantive is increasingly mediated through online discussion platforms. Newman et al. (2025) estimates that usage of social media as a source of news in the U.S. rose from 27% to 54% between the years 2013 and 2025, and McClain et al. (2024) shows that the most popular social media platform for political news is the discussion platform X. There is growing recognition that the incentive structures governing these platforms is a crucial factor determining whether conversations proceed in healthy or unhealthy directions (Aridor et al., 2024). Concerns about online discourse have become prevalent enough that governments around the world are exploring regulatory frameworks for moderating online content, with the EU’s Digital Services Act being a prominent example.

Most online discussion platforms are free to use. Some may charge for premium features, but these features are usually related to removing ads, content management tools, and features for gaining visibility or monetization. Few platforms charge a direct cost to post, and fewer still allow users to pay each other directly for content that they appreciate. Instead, the exchange of value between content creator and content consumer tends to be mediated through indirect mechanisms, such as ad revenue, visibility, and reputation—often referred to as “clout”—which may indirectly benefit the content creator’s endeavors outside the discussion platform proper. The indirect nature of these mechanisms may result in misaligned incentives between content creators and content consumers. For example, content creators may be more incentivized to create content that readers will *click*, rather than content that readers will actually read, digest, and ex-post appreciate the creator for producing.

There are a few reasons why online discussion platforms may not want to charge posting fees or allow users to pay each other. For one, economies of scale and network

effects suggest that platforms would want to onboard as many users with as little cost to the user as possible. But another, perhaps less appreciated reason is frictions in the payments infrastructure. The value of any individual post or comment is likely quite small, too small to cover the fixed fees for credit card transactions. Fee-less payments networks like Venmo do exist, but they usually have strict volume limits and charge fees to frequent users. Moreover, most popular discussion platforms, like X and Reddit, are international in scope. Routing payments between international users introduces multiple layers of frictions related to currency exchange and international banking regulations.

Despite this, there is good reason to believe that charging fees for posts and allowing users to pay each other directly could increase the quality of content posted. Signaling theory suggests that posting costs can be an effective deterrent against spam (Joseph and Thevaranjan, 2008). Empirically, Tchernichovski et al. (2019) showed that introducing a small fee to an online rating system filtered out low information signals and led to higher correlation between review ratings and quality. Empirical studies from other social media platforms have also shown that financial incentives in the form of revenue sharing can influence the quantity and quality of content (Sun and Zhu, 2013; El-Komboz et al., 2023; Kerkhof, 2024), though none of these studies investigated direct peer-to-peer payments and posting costs in online discussion platforms.

To our knowledge, there is little work to date about the impact of direct monetary incentives in online discussion platforms, despite its potential impact on the quality of online discourse. This paper aims to fill that gap using a unique dataset from an online discussion platform called Stacker News. Stacker News is an internet message board, in the style of Hacker News or Reddit, that uses Bitcoin micropayments sent over the Lightning Network. Users can send each other Bitcoin micropayments, called “zaps”

(usually on the order of 1-100 satoshis, but sometimes ranging into the thousands) to reward each other for posting content that they enjoy and to encourage the production of similar content. The platform also uses micropayments as a form of sybil, spam, and bot resistance. Users must pay a fee to post, which discourages them from posting low quality content or from creating sock puppet accounts. Users who consistently post high quality content can earn enough zaps to turn a profit net of posting fees. The micropayments on Stacker News are made possible by the low cost, instantaneous transaction speed, and international connectivity of the Lightning Network.

Because payments on Stacker News are made with real money (Bitcoin), it is an ideal setting for studying how financial micro-incentives affect internet discourse. There is variation in posting fees across time and across the platform’s various sub-forums (called territories). This group-level temporal variation allows us to study how content quality and content quantity change when posting costs change. Using a difference-in-differences strategy, we find that the quality and quantity of posts are indeed responsive to territory posting fees. The results indicate that when a territory doubles its posting cost, the expected number of weekly posts is reduced by 16.8%, but the expected quality of posts (as measured by zaps received in the first 48 hours) is increased by 16.4%. These results suggest that requiring users to pay to post can improve the quality of content posted. In addition to these findings, we also find that users learn what types of content other users like over time, and that over time this leads to an increase in average post quality. We also find that unprofitable users are more likely to leave the platform while profitable users are more likely to stay.

Our results suggest that equipping online discussion platforms with financial incentives in the form of posting costs and direct peer-to-peer micropayments can improve the quality of internet discourse. It also highlights a clear use case for the utility of Bitcoin and the Lightning Network, because the small, global, low-cost, and

instantaneous micropayments on Stacker News are not currently possible within the traditional banking system. This research therefore adds to the growing body of evidence showing how Bitcoin adoption enables new markets and influences real world outcomes—in this case, in the realm of internet discourse.

2 Incentive structures on Stacker News

From its outset, Stacker News was founded with the idea of using Bitcoin micropayments to incentivize quality content and good behavior. Its motto is “Stacker News is trying to fix online communities with economics.” There are a number of incentive structures built into the design of Stacker News.¹

Pay to post. First, users must pay to post. The posting cost is denominated in sats and varies by territory.² A territory is a sub-forum on Stacker News, usually topic-specific. As of October 2025, the most popular territories are `bitcoin`, which focuses on discussions about Bitcoin; `econ`, which focuses on news and discussion about economics; `Stacker_Sports`, which focuses on sports discussion; `AI`, focused on news and discussion about artificial intelligence; and `Politics_And_Law`, focused primarily on politics and current events.

Territories are created and operated by individual users on Stacker News. To create a territory, a user simply has to choose a name for the territory, write a description, and pay a monthly fee to Stacker News to maintain the territory. As of October 2025, the monthly fee to operate a territory is 50,000 sats.³ If a territory operator fails to pay the monthly cost, the territory becomes archived. Users are still

¹The following description of how the Stacker News works is accurate up to October 5, 2025.

²One sat (or Satoshi) is equal to 1/100,000,000th of a Bitcoin.

³Territory operators can also pay for 12 months at a time for 500,000 sats, or pay for a territory in perpetuity if they pay 3 million sats.

able to see posts in an archived territory, but they can no longer post in it. Any user (it doesn't have to be the territory founder) can unarchive a territory simply by choosing to pay the monthly fee, in which case they become the new operator.

Once a new territory is founded, users can post in that territory by paying a posting fee which is set by the territory operator. The territory operator earns 70% of the posting fee as revenue, and the other 30% goes to a daily rewards pool (more on that later). Territory operators are therefore incentivized to choose a posting fee that optimizes between post quantity, post quality, and revenue earned. If the posting fee is too low, the territory may not earn enough revenue to cover costs, and it may also attract low quality posts. On the other hand, if the fee is too high, users may be deterred from posting in the territory. Users are incentivized to post in the territory that best fits the topic of their post so as to maximize engagement and sats earned net of posting fees. Territory operators can additionally attract users to post in their territories by creating engaging content or by adding value to users' posts. For example, the operator of the `Stacker_Sports` territory often runs sports prediction games for users to engage in, and the operator of `BooksAndArticles` writes a weekly newsletter to highlight the territory's best posts, thus adding visibility and additional zaps for those creators.

Value for value. Second, users can tip each other directly for content that they like. In the Bitcoin community, systems that allow users to directly tip each other are called "value-for-value" systems. On Stacker News, these tips are called "zaps". To zap an item, the user clicks a lightning bolt icon which is featured prominently on every post and comment. When the user clicks the lightning bolt, sats are sent from the user to the creator of the item. The default zap amount is 100 sats (about 10 cents at the time of writing), but users can customize their own defaults. They can

also long-press the lightning bolt icon to zap a custom amount.

Like Reddit upvotes, zaps are used by the platform to surface high quality content—the default view on Stacker News ranks posts based on the amount of zaps earned by the post, modified by a time decay factor to keep the view fresh. Unlike Reddit upvotes, zaps represent real monetary value transferred from the zapper to the creator of the post. The zapping system incentivizes high quality content in two ways: not only does the system surface higher quality posts to the top of the home page, but it also lets creators of high quality content get paid directly by other users for their work.

While it is clear how zapping incentivizes content creators, it is not as clear what incentives users have to zap. Economists who have studied tipping in non-digital contexts have argued that tipping is not well explained by considerations of future value (Azar, 2020) and have instead focused on altruistic “warm glow” effects (Andreoni, 1990; Crumpler and Grossman, 2008). However, these are contexts where interactions are generally not repeated and where the payment includes a deterministic price along with an optional tip. On Stacker News, users have repeated interactions with each other and there is no fixed reward for content creators other than tips. The incentive model may therefore be closer to that of a “pay-what-you-want” system, in which tipping is rationally used to ensure that producers do not switch to a less advantageous pricing model for consumers (Mak et al., 2015). In the case of Stacker News, the consequence of not tipping could be a reduction in content production that the user enjoys. Whether users zap due to an altruistic desire to reward content that they enjoy, or by a rational incentive to support the continued production of such content, or by a desire to influence the ranking of posts, we do observe a healthy amount of zapping on Stacker News. We also find that the amount of zapping is correlated to objective measures of post quality.

Two additional features of zapping are worth mentioning. First, only 70% of the zap amount is received by the content creator. The remaining 30% is “taxed” and redistributed. 21% of the zap amount goes to the owner of the territory where the item was posted. Through this, territory owners are additionally incentivized to encourage high quality content in their territories. The remaining 9% of the zap goes to the daily rewards pool. This 30% “tax” on zaps is a form of sybil resistance: it discourages users from creating sock puppet accounts to zap their own content.

Daily rewards. Third, Stacker News incentivizes user behavior with a daily rewards system. As already mentioned, 30% of posting fees and 9% of zaps go to the daily rewards pool. There are three other main sources of daily rewards. First, users can donate directly to the rewards pool. This is not common, but some users take advantage of this as a way to support the platform. Second, users can post anonymously under a dedicated `anon` account. Any zaps earned by the `anon` account are sent to the rewards pool. Third, users can pay to boost their own posts. Boosting increases the post’s visibility based on the amount paid. When a user pays to boost a post, 70% of the amount goes to the owner of the territory that the post is in and 30% goes to the daily rewards pool.

Users are allocated sats from the daily rewards pool according a formula that depends on user behavior in that day. Users are rewarded for highly zapped posts and comments, as well as for zapping other highly zapped posts and comments early. The reward system amplifies the already present incentives to post good content, but it also incentivizes users to help identify and surface good content by zapping good content early.

Market-based content moderation. Lastly, Stacker News does not employ any content moderators, nor are there formal moderators in any territory other than the territory owners. Instead, Stacker News utilizes a form of market-based content moderation called “downzaps”. A downzap is like a zap, except the zapper pays to *downweight* the zapped item in any search or sort rankings. If enough users downzap an item, it becomes “outlawed” and will be hidden from view for most users.⁴

Getting downzapped or outlawed is not a good outcome for the item’s creator. It lowers the item’s visibility and makes it less likely that the post will accumulate zaps. Since posting requires fees, this makes it more likely that the poster will lose money on the post. Moreover, getting downzapped lowers a user’s trust score, which reduces their ability to earn rewards or influence other users’ post rankings.

The requirement for downzaps to be paid with sats is an important feature of this system. It prevents users from frivolously downzapping other users and it prevents users from spinning up sybil accounts to do the same. By requiring a monetary payment for downvoting, Stacker News ensures that only content which is truly objectionable gets downzapped.

Hypotheses. Based on the above discussion, we hypothesize the following about the incentive structures on Stacker News. First, we hypothesize that higher posting costs will lead to fewer posts in a territory, but higher quality posts. Second, we hypothesize that the types of content that are better rewarded by zaps will increase in prevalence over time, as users learn that those kinds of content are better rewarded. Third, we hypothesize that consistently unprofitable users are more likely to eventually leave the platform, whereas consistently profitable users are more likely to stay. We test these hypotheses one by one in the following sections of the paper.

⁴Users can opt in to seeing outlawed content in their settings.

3 Data

To test these hypotheses, we downloaded a snapshot of the Stacker News database on October 5, 2025. The data covers the time period from June 11, 2021 to October 5, 2025 and contains 207,219 posts and 914,266 comments (replies), for a total of 1,121,485 items posted by 11,228 users across 117 territories. Over this time period, users zapped each other 146 million sats and paid 13 million sats in posting fees.

In addition to the Stacker News data, we downloaded daily Bitcoin price data from CoinMarketCap.com. At the time of writing, the price of one Bitcoin is approximately US\$100,000, which would make 1,000 sats worth approximately \$1. This is a useful rule of thumb for converting between sats and dollars when trying to evaluate the economic significance of Stacker News micropayments.

Figure 1 shows the growth in Stacker News, as measured by weekly number of items posted, alongside the weekly Bitcoin price. In aggregate, both the Bitcoin price and the activity on Stacker News grew over this time period.

4 Pay-to-post and content quality

In this section, we consider our first hypothesis: whether higher posting costs lead to fewer posts in a territory, but higher quality posts. First, we develop a simple model of posting behavior to ground our empirical work. Then, we test whether the predictions of the model are supported by the data.

Model. We envision a model in which a user receives “post ideas” as a Poisson arrival process with rate γ . Post ideas arrive with a quality level q , which affects how other users respond to the post, and an intrinsic motivation to post it, ϵ . The level of internal motivation may be correlated with post quality, and they are drawn from

a joint probability density function $f(q, \epsilon)$. We allow different users to draw from different distributions of (q, ϵ) , but for now we focus on the decisions of a single user.

By Bayes' Rule, $f(q, \epsilon) = f_{\epsilon|q}(\epsilon|q)f_q(q)$, where $f_{\epsilon|q}(\cdot)$ is the conditional density of ϵ given q , and $f_q(\cdot)$ is the marginal density of q . We assume that $f_{\epsilon|q}(\cdot)$ is strictly log-concave and weakly satisfies the monotone likelihood ratio property in ϵ with respect to q :

Assumption 1 (Log Concavity). *For each q , $f_{\epsilon|q}(\cdot|q)$ is strictly log-concave; that is, $\ln f_{\epsilon|q}(\epsilon|q)$ is strictly concave in ϵ .*

Assumption 2 (Monotone Likelihood Ratio Property). *For any $q_2 > q_1$, the likelihood ratio of the conditional densities,*

$$\ell(\epsilon; q_1, q_2) = \frac{f_{\epsilon|q}(\epsilon|q_2)}{f_{\epsilon|q}(\epsilon|q_1)}$$

is weakly increasing in ϵ .

Remark. Assumption 1 limits the heaviness of the tail of the motivation distribution. Intuitively, it ensures that the extreme tail of the motivation distribution is not thick enough to swamp quality selection effects at very high posting costs. It is not a strong assumption, and it is satisfied by most commonly used distributional families, including the normal and logistic distributions.⁵

Assumption 2 is a non-negative dependence condition between post quality q and intrinsic motivation ϵ . Informally, it says that higher quality ideas tend to come with (weakly) higher intrinsic desire to post: when q is higher, the distribution of ϵ shifts to the right. Independence between ϵ and q is allowed as a special case, but the

⁵Weaker, local conditions about the hazard rate in the neighborhood of the current posting cost would also suffice, but we state the stronger condition of global log-concavity for simplicity. Moreover, strict log-concavity is assumed for simplicity, but it suffices to assume either strict log-concavity or strict monotone likelihood ratio property; we do not need strictness for both.

assumption rules out systematic negative dependence. In our context, this means that users who receive a high quality post idea are also (weakly) more likely to feel a strong intrinsic desire to post it. As we shall document later, this is consistent with users' self reported perceptions of their own motivations. $\square$

Once a post is made, zaps arrive according to a Poisson process. Both the arrival rate of zaps and the amount of each zap are dependent on the quality of the post, q . Let $\lambda(q)$ be the arrival rate as a function of q , and let $E[z|q]$ be the expected zap amount per zap conditional on q .

Supposing the user has a discount factor of ρ , the expected present value of the Poisson stream of zaps is equal to:⁶

$$V(q) = \frac{\lambda(q)}{\rho} E[z|q] \quad (1)$$

We assume that $V(q)$ is continuous and strictly increasing in q , which merely says that the expected present value of zaps is higher for higher quality posts:

Assumption 3 (Monotone Returns to Quality). $V(q) = \frac{\lambda(q)}{\rho} E[z|q]$ is continuous and strictly increasing in q .

The user will make the post if their expected present value of zaps plus their intrinsic motivation is greater than the posting cost c :

$$\text{Post} = 1 \text{ if } V(q) + \epsilon > c \text{ and } 0 \text{ otherwise} \quad (2)$$

From the perspective of the user, c is fixed and known at the time of the post.

⁶This follows from $V(q) = \int_0^\infty e^{-\rho t} \lambda(q) E[z|q] dt$.

We now demonstrate conditions under which an increase in posting cost c will lead to fewer posts, but higher quality posts.

Proposition 1. *Suppose Assumptions 1-3 hold. Then an increase in c :*

- (i) strictly decreases the probability that the post idea is posted, and*
- (ii) strictly increases the expected quality of posts, conditional on posting.*

Proof. A proof is provided in the appendix. □

Remark. Proposition 1 implies that when posting cost increases, users are less likely to post, but that when they do post, the average quality of the post is higher. Although the proposition is stated from the perspective of a single user, we note that as long as the global joint distribution of (q, ϵ) holds to Assumptions 1 and 2, and if Assumption 3 also holds for all users, then Proposition 1 applies in the aggregate, even if users have heterogeneous marginal densities for q and ϵ .

The result of Proposition 1 extends to a setting with multiple territories. Users post in the territory j that maximizes $V_j(q_j) + \epsilon_j - c_j$, provided it is positive. When territory j raises its posting cost c_j , users' outside options are unaffected, so the same logic applies: as long as the population-level joint distribution of quality and motivation within each territory satisfies Assumptions 1-3, a territory that raises its posting cost will see fewer but higher quality posts. □

Evidence on post quality. To empirically test the predictions of Proposition 1, we run regressions of the following form:

$$\ln \text{Quality}_{ijt} = \alpha \ln \text{Cost}_{ijt} + \delta_j + \gamma_t + \epsilon_{ijt} \quad (3)$$

where i indexes a post, j indexes a territory, and t indexes a year-week.⁷ Quality_{ijt} is the measure of post quality, Cost_{ijt} is the cost (in sats) of the post, δ_j is a set of dummy variables for each territory, and γ_t is a set of dummy variables for each week. The coefficient α identifies the elasticity between posting cost and post quality. We hypothesize that $\alpha > 0$.

To measure post quality, we measure the number of sats the post earned on zaps in the first 48 hours of posting. Per Assumption 3, this is strictly increasing in post quality and thus a valid proxy for the latent post quality. The purpose of the dummy variables is to absorb any unobserved heterogeneity in territories (e.g. if some territories are more popular and thus attract more zaps), and to absorb any global time trends in the popularity of Stacker News.

The causal effect of posting cost is identified from differential changes in post quality across territories when the territory posting costs change differentially. For example, if the posting cost in the `econ` territory increases while the posting cost in `bitcoin` stays the same, and subsequently the average quality of posts increases in `econ` relative to in `bitcoin`, this would suggest that the change in posting cause likely caused the change in post quality. The coefficient α is identified off the many such changes present in the data.

To give the reader a sense of the variation in posting costs, we plot the posting cost histories for four representative territories in Figure 2. The figure shows that there is indeed substantial variation in posting fees both across and within territories. In the data we use for the regression, the average posting cost is 51 sats, the average amount of sats earned in the first 48 hours is 364 sats, and the average number of comments in the first 48 hours is 3.3. There is substantial variation in all three: the standard deviation of the posting cost is 241 sats, of sats earned in the first 48 hours

⁷For example, 2024w1 is a different t than 2025w1.

is 6,662, and of comments in the first 48 hours is 9.6.

Before running the regression, we apply the following sample selection criteria. We do not include posts in the `AMA` territory or the `jobs` territory because these territories seem to operate on different incentive structures than the other ones.⁸ We also do not include a number of special post types that do not follow normal posting rules.⁹ Lastly, we do not include any posts made in territories by the owner of the territory. Since 70% of posting costs go to the territory owner, the incentives for territory owners to post in their own territories differ from that of non-owners.

Table 1 shows results for regressions of specification (3) where post quality is measured as zaps in the first 48 hours. Each column of Table 1 includes a different set of fixed effects in the regression. Column 1 includes no fixed effects, and is therefore a straight OLS regression of log zaps in the first 48 hours on log posting cost. Column 2 includes territory and week fixed effects to control for any heterogeneous effects between territories. Column 3 includes fixed effects for the identity of the territory owner to control for the zapping behavior of territory owners. For example, some territory owners zap every post that comes into their territory while others do not. Territory-owner fixed effects control for this behavior.¹⁰ Finally, column 4 includes user fixed effects to test whether the gains in post quality are driven by within-user effects or compositional effects.

Table 1 shows that there is a statistically significant and robust positive relationship between territory posting fees and post quality as measured by zaps in the first

⁸`AMA` is a territory where influential people can make “Ask Me Anything” posts. Because these posters are influential, posts in the `AMA` territory earn an unusually large amount of sats, and the posters are unlikely to be motivated by posting costs. `jobs` is a territory run by Stacker News in which companies can make job advertisements.

⁹These include profile posts, which users can post and edit for free, “saloon” posts, which are daily discussion threads posted by Stacker News, and freebies—a limited number of reduced-visibility posts that users are allowed to make for free.

¹⁰Note that territory fixed effects do not fully absorb territory owner fixed effects, since a single user can own multiple territories, and territories can also change owners over time.

48 hours. In terms of the magnitude of the effect, Table 1 column 3 (our preferred specification) suggests that the elasticity between territory posting cost and expected post quality is 0.219. This roughly translates to: if a territory doubles its posting cost, then the quality of posts is expected to increase by 16.4 percent. The relationship is not explained by territory specific effects, such as a spurious correlation between posting fees and the popularity of a territory’s topic, nor is it explained by global time trends in posting fees and zap behavior on Stacker News. Nor is it explained by effects related to territory owners—for example, if territory owners who set high fees also tend to zap posts in their territory more generously. The best explanation for the correlation between posting cost and post quality is that users will only choose to post in a higher cost territory if they think that the post is of high enough quality to warrant the cost. This is consistent with Proposition 1 and with users’ self reported perceptions of their own behavior on the platform. One Stacker News user is quoted saying:

“If I put a lot of effort into a post, I know that the posting fee will be offset by the zaps. And even if it doesn’t, I’m just happy to show the work I’ve done and I’m ready to pay for it. On the other hand, when I share a simple link with a few quick comments and quotes, it takes me a few minutes, and I look for the cheaper territory to post it. If I don’t find any cheap territory, I’d rather not post, as I don’t think paying to post for links is worth the same as paying to post for actual proof-of-work.”¹¹

It makes sense that users will only be willing to pay a high posting cost if they think the quality of the post justifies it—either because they will recuperate the

¹¹In this statement, the user is referring to link posts, which are posts that include a link to an external URL, often without any text in the post body. Link posts usually receive less zaps and are considered to be of lower quality because the content is not produced by the user themselves, and are thus less rewarded by zaps.

cost through zaps or because of a stronger intrinsic desire to post high quality work. However, one may wonder why users don't just choose a low cost territory to post in. There is no rule on Stacker News that says you must post content to the most relevant territory. Moreover, most territories on Stacker News are lightly moderated, so posting content to an irrelevant territory is not likely to result in post removal. Despite these caveats, in practice we observe that users tend to post in territories for which the content is most relevant. This may be for a few reasons. First, users can subscribe to or mute territories.¹² Thus, posting in a more relevant territory will increase the visibility of your post to users who are most likely to be interested in it. Second, territory owners sometimes add value to their territories by collecting, summarizing, and rewarding top posts. Your post is less likely to be highlighted if it is not relevant to the territory. Lastly, users may have an intrinsic, aesthetic desire to post in the most relevant territory, especially if they view their content to be of high quality.¹³

Finally, we comment on column 4 of Table 1, where we include user fixed effects. The inclusion of user fixed effects reduces the estimated elasticity between posting cost on post quality from 0.219 to 0.091 (still statistically significant at the 5% level). This suggests that the increase in post quality in response to an increase in posting cost is driven by both intensive margin (within-user) and extensive margin (compositional) effects. On the intensive margin, the same users now post higher quality posts, conditional on posting. On the extensive margin, the composition of posters in the territory shifts towards higher quality, most likely due to lower quality users dropping out. Column 3 remains our preferred specification over column 4, because we view the

¹²Subscribing to a territory means you will receive notifications when a new post is made to that territory. Muting a territory means that you will no longer see posts from that territory.

¹³In terms of the model, this would be reflected in a high positive affiliation between the territory specific components of (q_j, ϵ_j) .

effect on the composition of users as a dimension we want to capture when studying the relationship between posting cost and post quality.

Evidence on post quantity. Now we consider whether territory posting cost also influences the number of posts a territory receives. If users are sensitive to posting cost, then we expect that territory posting fee will be negatively correlated with number of posts. To test this, we run regressions of the following form:

$$\ln \text{Posts}_{jt} = \alpha \ln \text{Cost}_{jt} + \delta_j + \gamma_t + \varepsilon_{jt} \quad (4)$$

where j indexes a territory and t indexes a week. Posts_{jt} is the number of posts posted in territory j in week t , Cost_{jt} is territory j 's posting fee in week t (measured as the average across days), δ_j is a set of territory fixed effects, and γ_t is a set of week fixed effects. The results of this regression, for various combinations of fixed effects, is presented in Table 2. As expected, there is a robust and statistically significant negative relationship between territory posting cost and the quantity of posts in a territory. Using column 3 as our preferred specification, the results imply an elasticity of post quantity with respect to post cost of -0.265 . This translates to: if the posting cost doubles, the territory can expect to receive 16.8 percent fewer posts in a week.

Taken together, the results are both surprising and unsurprising. They are unsurprising in the sense that they conform with economic theory: demand curves slope downwards (when posting costs go up, number of posts goes down), and signaling theory works (when posting costs go up, higher quality posts are made). They are surprising in the sense that even such small micro-incentives (the average posting cost is just 51 sats, or about 5 cents) are enough to influence user behavior in such a way that post quality is improved. The results suggest that pay-to-post may be

an effective mechanism for mediating content quality, even at very small monetary amounts. Moreover, the assumptions required for this result to hold more broadly are fairly weak: we require only that expected rewards are increasing in post quality (Assumption 3), that post quality and intrinsic motivation to post are positively affiliated (Assumption 2), and that extreme levels of intrinsic motivation are sufficiently rare (Assumption 1). All three assumptions are likely to be satisfied in other social media environments, and thus it may be possible to extrapolate our results to other settings (though perhaps with different baseline cost and rewards levels.)

5 Value-for-value incentives and content quality

We now consider our second hypothesis: that the types of content better rewarded by zaps will increase in prevalence over time as users learn that these kinds of content are better rewarded. As in the previous section, we first extend the model of posting behavior to account for effort and learning, then we test the empirical predictions of the model.

Model. We extend the model from Section 4 to account for effort. As before, opportunities to post still arrive as a Poisson process with rate γ . But before observing (q, ϵ) , the user first decides how much effort to exert in crafting the post. Effort affects the distribution of post quality and intrinsic motivation, $f(q, \epsilon; e)$ (users may be more motivated to post if they’ve exerted a lot of effort). The cost of exerting effort is $\kappa(e)$ with $\kappa(0) = 0$ and $\kappa'(e) > 0$. The user’s expected utility conditional on effort is:

$$U(e) = E_{q, \epsilon | e} [\max \{V(q) - c + \epsilon, 0\}] - \kappa(e)$$

Without uncertainty, the model would be equivalent to the one presented in Section 4, just with the distribution of (q, ϵ) moderated by the optimal effort level. We therefore assume that users have uncertainty over the returns to quality. That is, let:

$$V(q) = \exp(\theta q) \tag{5}$$

where users have uncertainty over θ . Note that this is without loss of generality because q has no natural units.

Given that users are uncertain about θ , they must infer it from their received zaps relative to quality. We assume that users observe q . Let z_i be the zaps received within a fixed time period, e.g. 48 hours, for a specific post i . z_i is a noisy signal proportional to $V(q_i)$, and therefore:

$$\ln z_i = \theta + q_i + \text{noise} \tag{6}$$

Now let a user have a posting history of posts indexed by $i = 1, \dots, N$. An unbiased estimator for θ is:

$$\hat{\theta} = \frac{1}{N} \sum_{i=1}^N \ln z_i - q_i \tag{7}$$

That is, the user's unbiased estimate for θ is the average of their posts' zap "surprise", i.e. how much additional zapping it received over and above its baseline quality level.¹⁴

We can now demonstrate that effort exertion is increasing in $\hat{\theta}$ and that, as a corollary, the expected quality of the next post is increasing in $\hat{\theta}$.

Proposition 2. *Let $i = 1, \dots, N$ index a user's previously made posts. Let q_i denote*

¹⁴This assumes diffuse priors; if the user had a more precise prior, their posterior mean would still be linear in $\frac{1}{N} \sum \ln z_i - q_i$.

the quality of those posts, let $z_i \propto V(q_i)$ be a noisy signal proportional to the present value of zaps that post i receives, and let $\hat{\theta} = \frac{1}{N} \sum \ln z_i - q_i$. Furthermore, assume that $F(q|\epsilon; e)$ satisfies FOSD in e , conditional on ϵ . Then, optimal effort is weakly increasing in $\hat{\theta}$.

Proof. A proof is provided in the appendix. $\square$

Evidence of learning. Since $\hat{\theta}$ is the average of the zap surprises across previous posts, Proposition 2 suggests that we can test for the presence of learning by regressing the quality of a user’s next post on the average zap surprise for all prior posts made by that user. Specifically, let i index a user and let k index the user’s k th post. We would run regressions of the following form:

$$q_{i,k+1} = \beta_0 + \beta_1 \underbrace{\frac{1}{k} \sum_{j=1}^k [\ln z_{ij} - q_{ij}]}_{\hat{\theta}_{ik}} + \epsilon_{i,k+1} \quad (8)$$

where z_{ij} is the zaps user i received on their j th post after the first 48 hours, and q_{ij} is the quality of user i ’s j th post. If users are learning about the returns to effort from past zap surprises, we would expect to estimate $\beta_1 > 0$. Intuitively, identification comes from the following thought experiment: imagine two otherwise identical users, but by random chance user A has had a more positive zap history than user B for the same quality posts. User A therefore has a more optimistic view of the returns to effort than user B, and will on the subsequent post exert more effort, resulting in higher expected post quality.

To measure q_{ij} , we make use of equation (6), which is the idea that the zap amount within any fixed time window is a noisy measure of q_{ij} . We can therefore estimate q_{ij} for each post by regressing zaps on objectively measured post attributes, such as

number of words, whether the post is a link post, the number of images and links in the content, etc:

$$\ln z_{ij} = \theta + \underbrace{X_{ij}\beta}_{q_{ij}} + \varepsilon_{ij} \quad (9)$$

The model’s predicted values of $X_{ij}\hat{\beta}$ can then be used as unbiased estimates of the post quality q_{ij} .

Table 3 reports results from regressions based on equation (9). Column 1 starts with three attributes: the number of words in the post, whether or not the post is a link post, and the number of images and links in the post body. As discussed earlier, link posts are posts that point to a URL. They can have empty post bodies (no words), but the author of a link post can also add text to the post body to provide context for the link. As confirmed by the results in Table 3, link posts are generally less well-rewarded on Stacker News than non-link posts. By contrast, posts with links and images *in the post body* are usually better rewarded on average than posts without links and images. The presence of images and links in the post body may signal that the author has spent some time thinking about and working on the post, both to make it more attractive and engaging (images), and to make it more useful (referrals to other relevant URLs).

Column 2 of Table 3 adds text embeddings to the vector of post attributes. The embeddings were computed from post text using OpenAI’s `text-embedding-3-small` embedding model. The embedding model computes a 1,536 dimensional embedding vector for each post that encodes the meaning of the text in such a way that two posts with similar text will have short distances between their embedding vectors. Before including these embedding vectors in the vector of attributes, we first reduce their dimensionality by performing principal components analysis (PCA) with 10

components. The PCA extracts the 10 orthogonal component vectors that explain the most variance in the original 1,536 dimensional vector. We then include the 10 principal components in the attribute vector X_{ij} for the regression.¹⁵

Lastly, Column 3 of Table 3 includes user fixed effects and territory fixed effects. It is appropriate to include these features in our definition of q_{ij} , because q_{ij} is meant to capture all the predictable features of post quality, and it is reasonable to assume that users have a sense of which territories are better rewarded and a sense of the baseline quality of their own writing.

Column 3 of Table 3 is our preferred specification. We therefore take the coefficients as estimated in Column 3 of Table 3 (including the user and territory fixed effects), and estimate q_{ij} for each post using only those attributes. We then regress $q_{i,k+1}$ on $\hat{\theta}_{ik}$, the average of past post residuals, as suggested in equation (8). Table 4 shows the results of this regression. Note that user and territory fixed effects are included in the regression by default. This is necessary because these fixed effects are mechanically correlated with $q_{i,k+1}$ and because we are trying to control for two otherwise identical users who have different histories of zap surprises. In Column 2 of Table 4, we additionally include the user’s history of total prior zaps and total prior to posts, to further control for time-varying user heterogeneity. Lastly, in Column 3 we include week fixed effects to control for global time trends in the quality of posts. Global time trends could itself be due to learning: especially if users learn not just from their own posts but from the posts of other users. Controlling for global time trends therefore isolates the effect of β_1 to the individualized learning that a user experiences from their personal zap history.

Using Column 3 of Table 4 as our preferred specification, the results show an elasticity of 0.024 between subsequent post quality and the average of past zap sur-

¹⁵Please refer to the appendix for further methodological details.

prises. The effect is modest: it implies that if user A has twice the average past zap surprise of user B, that user A’s subsequent post will be of 1.7 percent higher quality, measured by expected zaps received. Although the magnitude of the effect is small, the effect is statistically significant and robust to the inclusion of various controls, as shown in Table 4.

Moreover, we note that this regression estimates learning based on individual experiences only, and partials out the effect of global learning via the week fixed effects. If we hypothesize that global learning is also present, then we should expect to see higher post quality over time, as measured by objective post attributes. And this is indeed what we see: Figure 3 plots the average of q_{ij} by week, showing that this has indeed been increasing over time. In other words, post quality—as measured by number of words and semantic features—has been increasing over time, which is evidence of global learning and quality improvement driven by value-for-value incentives.¹⁶

6 Profitability and user retention

We now test our last hypothesis: that consistently unprofitable users are more likely to exit the platform, whereas profitable users are more likely to stay. This would arise straightforwardly out of the learning model in which some users learn that their returns to effort are consistently low.

To measure profitability, we only consider whether the user is able to earn enough zaps and rewards to cover their posting costs. We do not consider their own zaps to other users as part of their costs, since zapping is entirely optional. Moreover, some users may have altruistic motives to zap, as discussed in the introduction, so greater zapping could indicate a greater commitment to the platform.

¹⁶Note that this is explicitly *not* driven by changes to zap generosity over time, since q_{ij} is measured based on post characteristics only.

We define a user as inactive if they go four or more consecutive weeks without making any posts, any comments, or zapping any items. The first week of inactivity is considered the week they went inactive. Inactive users can become active again, though we treat this event as exogenous.

The goal of this section is to estimate how user profitability affects the probability of becoming inactive. We estimate linear probability models of the following form:

$$\text{Inactive}_{it} = \alpha \text{Unprofitable}_{it} + \beta \text{Unprofitable}_{it} \times \text{Growth}_t + \gamma_t + \epsilon_{it} \quad (10)$$

where i indexes a user and t indexes a week. Inactive_{it} is a binary indicator for whether user i became inactive on week t , Unprofitable_{it} is a binary indicator for whether the user was unprofitable in the last 8 weeks, and Growth_t is the percentage growth in Bitcoin-USD price in the last 8 weeks. γ_t is a week fixed effect to capture global time trends, and ϵ_{it} is the error term. We hypothesize that $\alpha > 0$, indicating that users who have been unprofitable are more likely to become inactive, and that $\beta > 0$, that this effect is amplified when Bitcoin’s price has recently appreciated.

Table 5 reports the results for this regression. Column 1 shows the result when only profitability in the last 8 weeks is included as a regressor. Column 2 additionally controls for the number of items the user posted in the last 8 weeks. Unsurprisingly, users that have posted more items in the last 8 weeks are less likely to become inactive, as it shows a greater commitment to the platform. Including the number of items in the last 8 weeks significantly moderates the effect of unprofitability, suggesting that there are some unprofitable users who post a lot anyway. Column 3 adds week fixed effects to control for global time trends, but the coefficient of interest is not much changed. Lastly, column 4 interacts unprofitability with the Bitcoin price appreciation over the last 8 weeks.

Using column 4 as the preferred specification, the results show that users who were unprofitable in the last 8 weeks are 5.5 percentage points more likely to become inactive than users who were profitable in the last 8 weeks. This is in comparison to a baseline inactive probability of 14.1% per week. In addition, column 4 shows that unprofitable are even more likely to become inactive when the Bitcoin price has recently appreciated. If the Bitcoin price has appreciated by 10 percent in the last 8 weeks, this increases the effect of unprofitability from 5.5 percentage points to 6.1 percentage points increased probability of exit. This suggests that users are sensitive to not just their profitability as measured in sats, but also their profitability as measured in purchasing power.

7 Conclusion

Our analysis of Stacker News data confirms three central hypotheses: first, that posting costs can increase the quality of posted content, consistent with signaling theory; second, that the potential to be rewarded with tips also leads users to post better content over time; and third, that unprofitable users are more likely to leave the platform while profitable users are more likely to stay. This third point suggests that content quality may further increase over time as poor quality users, such as bots or trolls, eventually leave the platform.

More broadly speaking, the results suggest that users of internet discussion platforms are responsive to financial micro-incentives, even ones of very small monetary value. Equipping a discussion platform with the proper incentive structures may therefore be an effective method to improve the quality of content and discourse on the platform, leading to fewer poor quality posts, more original content, and more respectful content.

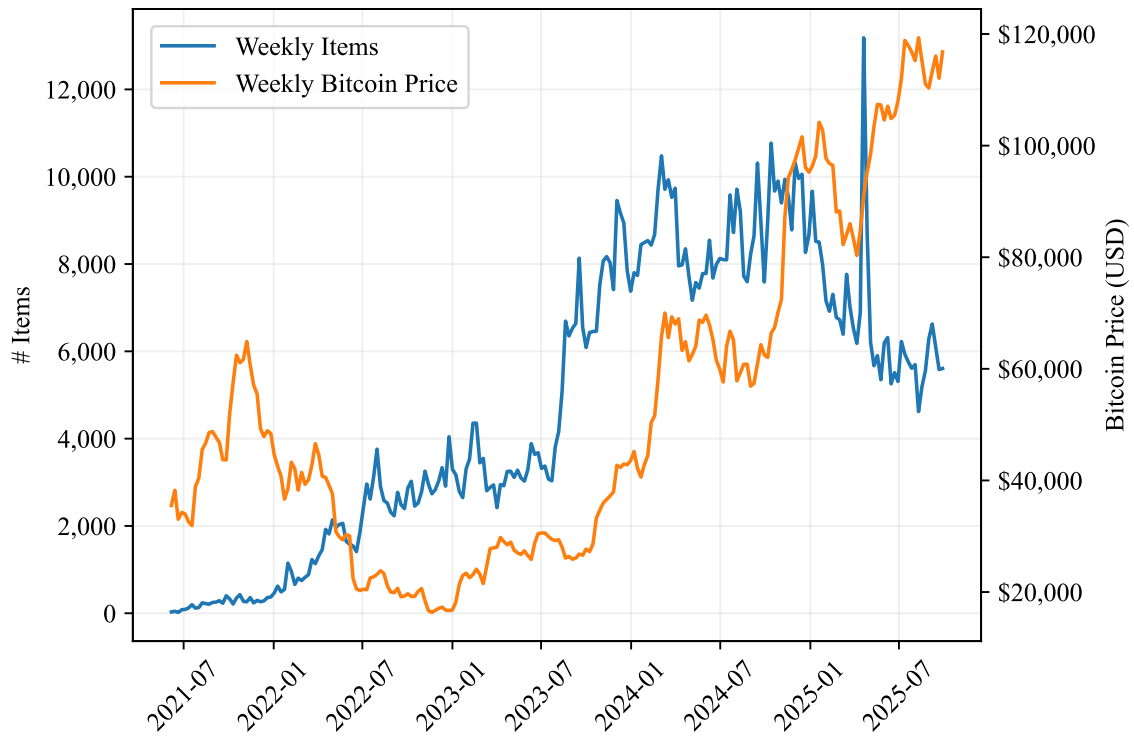
Moreover, the incentive structures on Stacker News are only made possible by the ability of the Lightning Network to instantaneously process small micropayments across a global network of users. This highlights a very clear and compelling use case for the utility of Bitcoin and the Lightning Network.

References

- Andreoni, J. (1990). Impure altruism and donations to public goods: A theory of warm-glow giving. *The Economic Journal* 100(401), 464–477.
- Aridor, G., R. Jiménez-Durán, R. Levy, and L. Song (2024). The economics of social media. *Journal of Economic Literature* 62(4), 1422–1474.
- Azar, O. H. (2020). The economics of tipping. *Journal of Economic Perspectives* 34(2), 215–236.
- Crumpler, H. and P. J. Grossman (2008). An experimental test of warm glow giving. *Journal of Public Economics* 92(5), 1011–1021.
- El-Komboz, L. A., A. Kerkhof, and J. Loh (2023). Platform partnership programs and content supply: Evidence from the Youtube “Adpocalypse”. CESifo working paper no. 10363.
- Joseph, K. and A. Thevaranjan (2008). Investigating pricing solutions to combat spam: Postage stamp and bonded senders. *Journal of Interactive Marketing* 22(1), 21–35.
- Kerkhof, A. (2024). Advertising and content differentiation: Evidence from youtube. *The Economic Journal* 134(663), 2912–2950.

- Mak, V., R. Zwick, A. R. Rao, and J. A. Pattaratanakun (2015). “Pay what you want” as threshold public good provision. *Organizational Behavior and Human Decision Processes* 127, 30–43.
- McClain, C., M. Anderson, and R. Gelles-Watnick (2024). How Americans navigate politics on TikTok, X, Facebook and Instagram. Report, Pew Research Center.
- Newman, N., A. R. Arguedas, C. T. Robertson, R. K. Nielsen, and R. Fletcher (2025). Digital news report 2025. Report, Reuters Institute for the Study of Journalism, University of Oxford.
- Sun, M. and F. Zhu (2013). Ad revenue and content commercialization: Evidence from blogs. *Management Science* 59(10), 2314–2331.
- Tchernichovski, O., L. C. Parra, D. Fimiarz, A. Lotem, and D. Conley (2019). Crowd wisdom enhanced by costly signaling in a virtual rating system. *Proceedings of the National Academy of Sciences* 116(15), 7256–7265.

Figure 1: Weekly Stacker News Items and Bitcoin Price



Note: Shows the number of items posted on Stacker News by week along with the weekly Bitcoin price. Weekly Bitcoin price is calculated as the average of the midpoint between daily highs and lows as reported by CoinMarketCap.com.

Figure 2: Posting Cost Histories for Select Territories

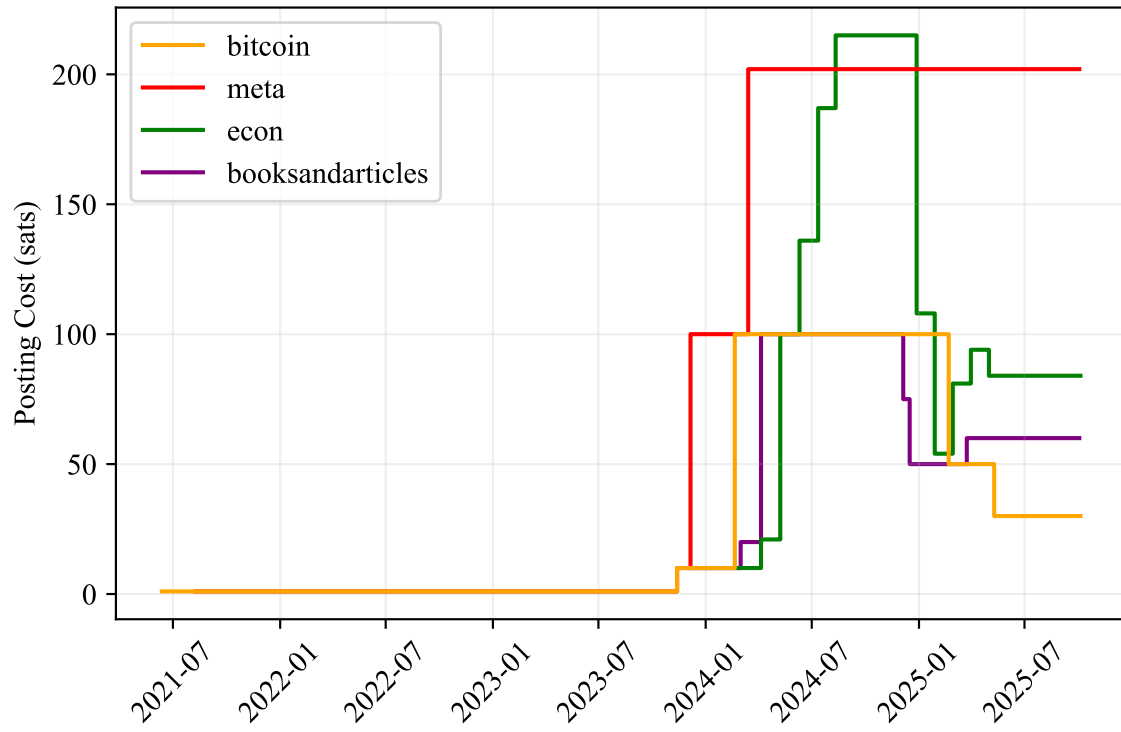
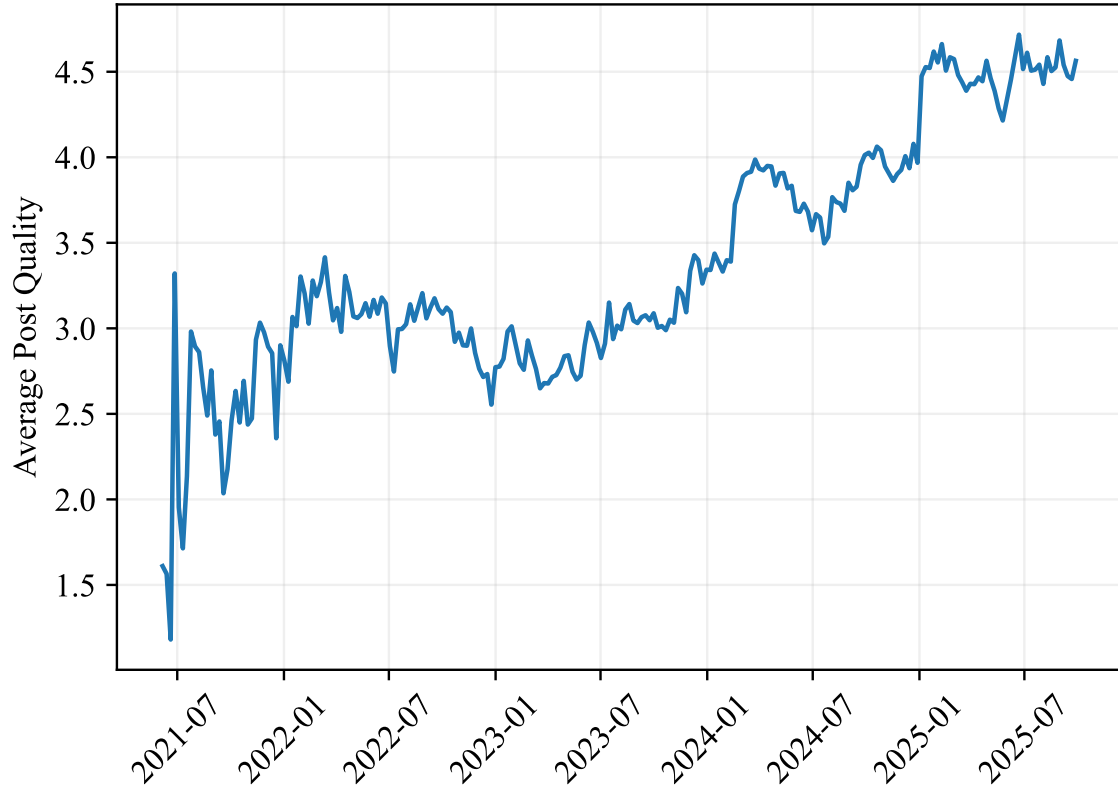


Figure 3: Post Quality Over Time



Note: Shows the average post quality as measured by objective post attributes (q_{ij}) for each week. See Section 5 for details.

Table 1: Effect of Posting Cost on Post Quality

	<i>Dependent variable:</i>			
	log(Zaps)			
	(1)	(2)	(3)	(4)
log(Posting Cost)	0.393*** (0.048)	0.273*** (0.058)	0.219*** (0.050)	0.091** (0.044)
Constant	2.530*** (0.076)			
Territory FE	N	Y	Y	Y
Week FE	N	Y	Y	Y
Territory Operator FE	N	N	Y	Y
User FE	N	N	N	Y
Observations	179,480	179,477	179,476	178,197

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Regression of number of sats a post earns in the first 48 hours on posting cost. The unit of observation is a post. Standard errors are clustered by territory.

Table 2: Effect of Posting Cost on Post Quality

	<i>Dependent variable:</i>		
	log(Posts)		
	(1)	(2)	(3)
log(Posting Cost)	-0.242*** (0.079)	-0.266*** (0.063)	-0.265*** (0.074)
Constant	3.328*** (0.415)		
Territory FE	N	Y	Y
Week FE	N	Y	Y
Territory Operator FE	N	N	Y
Observations	6,023	6,020	6,020

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Regression of number of posts in a territory on posting costs for that week. The unit of observation is a territory-week. Standard errors are clustered by territory.

Table 3: Regression of Post Zaps on Post Attributes

	<i>Dependent variable:</i>		
	log(Zaps)		
	(1)	(2)	(3)
log(# Words)	0.322*** (0.034)	0.254** (0.101)	0.161*** (0.044)
Is link post	-0.485** (0.239)	-0.197** (0.094)	-0.404*** (0.058)
# Images and links in post body	0.020* (0.012)	0.015** (0.007)	0.011* (0.006)
Constant	3.253*** (0.314)	3.200*** (0.341)	
Text Embeddings	N	Y	Y
Territory FE	N	N	Y
User FE	N	N	Y
Observations	175,857	175,857	174,578
R^2	0.155	0.180	0.404

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Regression of number of zaps a post earns in the first 48 hours on post attributes. The unit of observation is a post. Standard errors are clustered by territory.

Table 4: Regression of Next Post Quality on Past Zap Surprises

	<i>Dependent variable:</i>		
	Quality of next post ($q_{i,k+1}$)		
	(1)	(2)	(3)
Avg of past zap surprises ($\hat{\theta}_{ik}$)	0.048*** (0.007)	0.034*** (0.008)	0.024*** (0.006)
log(Total prior zaps)		0.016*** (0.004)	-0.012*** (0.003)
log(Total prior posts)		0.000*** (0.000)	-0.000 (0.000)
Territory FE	Y	Y	Y
User FE	Y	Y	Y
Week FE	N	N	Y
Observations	170,936	170,936	170,936

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Regression of next post quality ($q_{i,k+1}$) on the average of prior zap surprises ($\hat{\theta}_{ik} = \frac{1}{k} \sum_{j=1}^k \ln z_j - q_j$). The unit of observation is a post. Standard errors are clustered by territory.

Table 5: Effect of Profitability on User Exit

	<i>Dependent variable:</i>			
	Whether user becomes inactive			
	(1)	(2)	(3)	(4)
Unprofitable in last 8 weeks	0.105*** (0.003)	0.062*** (0.003)	0.060*** (0.003)	0.055*** (0.003)
$\dots \times$ BTC price appreciation in last 8 weeks				0.056*** (0.014)
$\log(\text{Items posted in last 8 weeks})$		-0.029*** (0.001)	-0.029*** (0.001)	-0.029*** (0.001)
Constant	0.079*** (0.001)	0.157*** (0.002)		
Week FE	N	N	Y	Y
Observations	92,956	92,956	92,956	92,956

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Linear probability model of whether a user becomes inactive in a given week. A user becomes inactive if they start a greater than four week consecutive period of no activity. The unit of observation is a user-week.

Online Appendix

A Proofs

Proof of Proposition 1

Proof. A post idea (q, ϵ) is posted if and only if $\epsilon > c - V(q)$. Define $\pi(q, c)$ as the probability (over ϵ) that a post of quality is posted when the cost is c :

$$\pi(q, c) = P(\epsilon > c - V(q))$$

$\pi(q, c)$ is equivalent to the survival function of $\epsilon|q$, evaluated at $c - V(q)$: $\pi(q, c) = S_{\epsilon|q}(c - V(q)|q)$.

(i) The unconditional probability of posting is:

$$P(\text{post}) = \int S_{\epsilon|q}(c - V(q)|q) f_q(q) dq$$

Since $S_{\epsilon|q}(\cdot|q)$ is a survival function, its first derivative is negative everywhere. An increase in c will therefore reduce the value of $S_{\epsilon|q}(c - V(q))$ at every q . Since that is what we are integrating over to get $P(\text{post})$, this strictly reduces $P(\text{post})$. The expected number of posts per unit time is $\gamma P(\text{post})$, so an increase in c strictly reduces the expected number of posts.

(ii) Take any $c' > c$. The set of ideas that were posted at cost c can be partitioned into two groups:

- **Surviving posts.** Ideas that are still posted at cost c' , i.e. $V(q) + \epsilon > c'$
- **Marginal posts.** Ideas that are posted at c but not at c' , i.e. $c < V(q) + \epsilon \leq c'$

$$V(q) + \epsilon < c'.$$

By the Law of Iterated Expectations,

$$E[q|\text{post at } c] = \alpha E[q|\text{surviving}] + (1 - \alpha)E[q|\text{marginal}]$$

where $\alpha = P(\text{surviving}|\text{post at } c)$. Noting that $E[q|\text{post at } c'] = E[q|\text{surviving}]$, we rearrange the above equation to obtain:

$$E[q|\text{post at } c'] - E[q|\text{post at } c] = (1 - \alpha) (E[q|\text{surviving}] - E[q|\text{marginal}])$$

Thus, to show that expected post quality is increasing in c , it suffices to show that the expected quality of the surviving posts is always higher than the expected quality of the marginal posts.

To demonstrate this, it is sufficient to establish that for any $q_2 > q_1$, the ratio $\pi(q_2, c)/\pi(q_1, c)$ is strictly increasing in c . Intuitively, this says that as costs rise, higher quality posts retain a larger share of the posting probability relative to lower quality ideas. This would imply that the marginal posts are tilted towards lower quality than surviving posts.

We write:

$$\frac{\pi(q_2, c)}{\pi(q_1, c)} = \frac{S_{\epsilon|q}(c - V(q_2)|q_2)}{S_{\epsilon|q}(c - V(q_1)|q_1)}$$

Define the hazard rate: $h(\epsilon|q) = f_{\epsilon|q}(\epsilon|q)/S_{\epsilon|q}(\epsilon|q)$. Differentiating $\pi(q_2, c)/\pi(q_1, c)$ with respect to c shows us that the ratio is increasing in c if and only if:

$$h(c - V(q_1)|q_1) > h(c - V(q_2)|q_2)$$

The proof now proceeds in two parts.

Claim 1. First, we show that $h(\epsilon|q_1) \geq h(\epsilon|q_2)$ for any $q_2 > q_1$. By the MLRP, for any $t > \epsilon$, we have that $\ell(t; q_1, q_2) \geq \ell(\epsilon; q_1, q_2)$, which rearranges to:

$$f_{\epsilon|q}(t|q_2)f_{\epsilon|q}(\epsilon|q_1) \geq f_{\epsilon|q}(\epsilon|q_2)f_{\epsilon|q}(t|q_1)$$

Integrating both sides over t from ϵ to ∞ gives us:

$$S_{\epsilon|q}(\epsilon|q_2)f_{\epsilon|q}(\epsilon|q_1) \geq S_{\epsilon|q}(\epsilon|q_1)f_{\epsilon|q}(\epsilon|q_2)$$

which rearranges to $h(\epsilon|q_1) \geq h(\epsilon|q_2)$ as was to be demonstrated.

Claim 2. Now, we show that $h(\epsilon|q)$ is increasing in ϵ for each q . Take the derivative of $h(\epsilon|q)$ with respect to ϵ . The sign is equal to the sign of:

$$f_{\epsilon|q}(\epsilon|q)^2 + f'_{\epsilon|q}(\epsilon|q)S_{\epsilon|q}(\epsilon|q)$$

When $f'_{\epsilon|q}(\epsilon|q) \geq 0$, this is immediately positive. When $f'_{\epsilon|q}(\epsilon|q) < 0$, we rely on the log-concavity property. Since $f_{\epsilon|q}(\cdot)$ is log-concave, the following holds for every $t > \epsilon$:

$$\ln f_{\epsilon|q}(t|q) < \ln f_{\epsilon|q}(\epsilon|q) + (t - \epsilon) \frac{f'_{\epsilon|q}(\epsilon|q)}{f_{\epsilon|q}(\epsilon|q)}$$

Exponentiating both sides:

$$f_{\epsilon|q}(t|q) < f_{\epsilon|q}(\epsilon|q) \exp \left[(t - \epsilon) \frac{f'_{\epsilon|q}(\epsilon|q)}{f_{\epsilon|q}(\epsilon|q)} \right]$$

Now integrating over t from ϵ to ∞ :

$$\begin{aligned} S_{\epsilon|q}(\epsilon|q) &< f_{\epsilon|q}(\epsilon|q) \int_{\epsilon}^{\infty} \exp \left[(t - \epsilon) \frac{f'_{\epsilon|q}(\epsilon|q)}{f_{\epsilon|q}(\epsilon|q)} \right] dt \\ &= - \frac{f_{\epsilon|q}(\epsilon|q)^2}{f'_{\epsilon|q}(\epsilon|q)} \end{aligned}$$

Since $f'_{\epsilon|q}(\epsilon|q) < 0$, this can be rearranged to:

$$f_{\epsilon|q}(\epsilon|q)^2 + f'_{\epsilon|q}(\epsilon|q) S_{\epsilon|q}(\epsilon|q) > 0$$

which was to be demonstrated. Thus, $h(\epsilon|q)$ is strictly increasing in ϵ for all q .

By the monotonicity assumption, $c - V(q_1) > c - V(q_2)$. By Claim 2, this implies that $h(c - V(q_1)|q_1) > h(c - V(q_2)|q_1)$. By Claim 1, $h(c - V(q_2)|q_1) \geq h(c - V(q_2)|q_2)$. Together, we have $h(c - V(q_1)|q_1) > h(c - V(q_2)|q_2)$, which was to be demonstrated.

□

Proof of Proposition 2

Proof. Given $\hat{\theta}$, the user's expected utility from effort is:

$$U(e; \hat{\theta}) = E_{q, \epsilon|e} \left[\max \left\{ \exp(\hat{\theta}q) - c + \epsilon, 0 \right\} \right] - \kappa(e)$$

It suffices to show that $U(e; \hat{\theta})$ is supermodular in $(e, \hat{\theta})$. For any $\hat{\theta}_2 > \hat{\theta}_1$:

$$\begin{aligned} U(e; \hat{\theta}_2) - U(e; \hat{\theta}_1) &= \\ E_{q, \epsilon|e} \left[\max \left\{ \exp(\hat{\theta}_2 q) - c + \epsilon, 0 \right\} - \max \left\{ \exp(\hat{\theta}_1 q) - c + \epsilon, 0 \right\} \right] \end{aligned}$$

The integrand is supermodular in $(\hat{\theta}, q)$ because $\exp(\hat{\theta}q)$ is supermodular in $(\hat{\theta}, q)$ and because the max operator is convex and increasing. Furthermore, since $F(q|\epsilon; e)$ satisfies FOSD in e , it follows immediately $U(e; \hat{\theta}_2) - U(e; \hat{\theta}_1)$ is increasing in e , and hence $U(e; \hat{\theta})$ is supermodular in $(e, \hat{\theta})$. $\square$